Beyond Reconstruction: A Real2Sim2Real Harness for Robot Policy Improvement

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Abstract: Can foundation-model-generated simulations improve real-world robot policies? We present an end-to-end Real2Sim2Real framework combining agentic scene reconstruction, interaction-aware asset completion, and simulation-based policy adaptation. Our harness orchestrates pretrained perception, reasoning, and 3D generation models with geometric and physical validation to construct diverse, interactive manipulation environments. We use frozen robot policies as behavioral discriminators to identify and correct real–sim discrepancies before training residual controllers in simulation. Evaluated across two independently fine-tuned $\pi_{0.5}$ policies, distinct data-collection paradigms, and manipulation tasks, our approach improves real-world camera-pickup success from 2/20 to 13/20 and plug-insertion success from 6/20 to 11/20, with corresponding gains in simulation. These preliminary results demonstrate the potential of validated, policy-aligned simulations for improving real-world manipulation. Project website: https://genrobo.github.io/r2s2r/.

Keywords: Real2Sim2Real, foundation models, policy evaluation

1 Introduction

Foundation models have made image-to-3D reconstruction increasingly accessible [1], yet turning these reconstructions into useful robot learning environments remains challenging. Static model pipelines propagate errors in segmentation, geometry, and object placement without validating their physical consequences, while generated assets often lack the articulation and contact mechanics required for manipulation. More fundamentally, even a geometrically plausible simulator may elicit different behavior from a pretrained robot policy due to mismatches in camera observations, robot embodiment, or state representation. This creates a critical dependency for Real2Sim2Real [2]: before using simulation to improve a deployed policy, we must first establish that the simulator reproduces its behavior under corresponding real-world conditions [3]. We address these challenges through an agent-driven reconstruction and validation harness, interaction-aware asset completion, and frozen-policy-guided simulator alignment, enabling subsequent policy adaptation through simulated experience.

2 Approach

1. Geometric and physical validity: Agentic real-to-sim reconstruction. Building on SimFoundry [4], we develop an agent-driven reconstruction harness that integrates Depth Anything V3 [5] for metric depth, SAM3 [6] for segmentation, Hunyuan3D 2.1 [7] for geometry and appearance, FoundationPose [8] for object registration, and GPT-6 Astra [9] for visual reasoning and occlusion completion. Unlike static model pipelines, Claude orchestrates twelve independently checkpointed stages through an MCP [10, 11] server, inspecting intermediate outputs and enforcing five

review gates with operator input. Automated checks validate object completeness, image registration, scale consistency, collision penetration, and physical settling, enabling early error detection and recovery from verified checkpoints.

2. Interaction validity: From reconstructed meshes to functional mechanisms. Since image-based reconstruction does not reliably recover hidden articulation or contact geometry, we introduce an agent-driven interaction-completion stage. Given natural-language descriptions of object functionality, the agent generates structured mechanism specifications that fit parametric hinges, socket cavities, or threaded contacts to reconstructed assets while preserving their appearance. Physical properties are initialized from geometric estimates and supplied priors, and completed mechanisms are evaluated through interaction-specific physics probes. These tests establish functional behavior of the authored mechanisms, rather than identification of their real-world physical parameters.

3. Behavioral fidelity: Policy-guided simulator alignment and adaptation. Geometrically and physically valid simulations may still induce incorrect policy behavior due to discrepancies in rendering, camera calibration, embodiment, or state representation. We therefore use the frozen pretrained $\pi_{0.5}$ [12] policy itself as a task-aware discriminator of real–sim fidelity. At corresponding real and simulated states, we compare action predictions under matched proprioception and task instructions, using behavioral discrepancies to guide targeted simulator corrections. Crucially, our objective is not to maximize policy success in simulation, but to reproduce its real-world behavior, including failure modes. Once the simulator is behaviorally aligned, we freeze the base policy and learn a lightweight residual controller from simulated oracle demonstrations, predicting bounded end-effector corrections during contact-rich insertion. This separates *simulator alignment* from *policy improvement*: the former aims to reproduce real-world policy behavior, while the latter uses the reconstructed environment to improve it without retraining the foundation policy.

3 Experimental Results

Reconstruction and interaction validity. We demonstrate our harness across diverse real-world environments, including multi-object tabletop scenes, desktop workspaces, robot manipulation workcells, and contact-rich objects such as power strips, cardboard boxes, and threaded jars. Beyond visual reconstruction, these environments support physical interactions spanning object pickup, articulated box folding, plug insertion, and bimanual jar opening. The harness detects and rejects geometrically or physically inconsistent reconstructions, while interaction-specific probes validate the completed mechanisms.

Policy-guided simulator alignment. Despite passing geometric and physical validation, our initial simulator failed to reproduce the pretrained policy's real-world behavior: the policy initiated manipulation in only half of simulated observations, compared to all real observations. Using the frozen policy as a behavioral discriminator, we systematically identified and corrected discrepancies in camera appearance, viewpoint, robot embodiment, and state feedback. Appearance corrections restored action initiation from 6/12 to 12/12 trials, while geometric calibration eliminated systematic grasp biases of 2.5–4 cm. These corrections enabled the previously unreliable policy to consistently grasp (12/12) and lift (10/12) objects in simulation. Transport and insertion remained unreliable, motivating residual policy adaptation.

Simulation-based residual policy adaptation. We evaluate our Real2Sim2Real framework across two distinct manipulation tasks, data-collection paradigms, and independently fine-tuned $\pi_{0.5}$ checkpoints. For plug insertion, demonstrations are collected using a GELLO-style teleoperation device [13], while camera pickup and placement uses a UMI-style handheld data-collection system [14]. For plug insertion, a gated planar correction is trained on 80 simulated episodes to compensate for residual alignment errors. Across 24 unseen object placements, the pickup residual improves grasp-and-hold success from 2/24 to 14/24 in simulation, with corresponding real-world improvement from 2/20 to 13/20. For plug insertion, success increases from 7/20 to 13/20 in simulation and from 6/20 to 11/20 on hardware. Detailed evaluations, ablations, and robot demonstrations are available on the project website.

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